



HELLIER South Central

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A Rockwood Company

Basic Level III Refresher Course Examination

Serial Number: Basic-Level III-17054

This is a closed-book examination to be completed without reference to any material other than that included in the examination package, or allowed by the examination monitor. Read each question and the responses carefully. Each question has only one correct answer; if you feel that more than one response is correct you must determine which answer best answers the question. After you have selected your answer circle the appropriate letter. If you decide to change your answer, mark an X through the incorrect answer, initial that change, and circle the correct answer. If you have more than one answer marked, that question will be counted as incorrect.

Examination Reviewed and Approved by: Eugene Charpia Date 5/19/2017
Revision A made to modify questions 98 & 109 -- Eugene Charpia 12/08/18

BASIC LEVEL III REFRESHER EXAMINATION

SECTION I

Interpretation of SNT-TC-1A and CP-189, 2016 Edition

1. If an employer chooses to increase the number of questions for a Specific examination over the recommended number in SNT-TC-1A, which of the following meets the recommendations?
 - a. ASNT requirements are mandatory and the number of questions cannot be changed.
 - b. Only an ASNT approved written practice shall be acceptable when the recommendations of SNT-TC-1A are not met.
 - c. The employer is free to modify the number of questions only in the General examination and must maintain the exact number of questions in the Specific examination.
 - d. The employer's written practice should state the number of questions that are required.
 - e. The number of questions may be increased only if there are more than two procedures encountered during specific assignments.
2. Which of the following is not considered an NDT method in SNT-TC-1A?
 - a. Visual testing
 - b. Thermal / infrared testing
 - c. Eddy current testing
 - d. Acoustic emission testing
 - e. Ground penetrating radar
3. ASNT intended that the requirements of CP-189 be applied:
 - a. with flexibility and reason.
 - b. as modified by other codes or standards.
 - c. as minimum requirements.
 - d. as maximum requirements.
4. The number of training hours recommended by SNT-TC-1A for Levels I and II:
 - a. should be increased by 25% when a person fails an end-of-course exam.
 - b. may be adjusted to reflect the candidate's actual education level, as described in the employer's written practice.
 - c. may not be altered from those published.
 - d. are increased by 25 percent for those persons not having a high school diploma or GED.
 - e. are reduced by 25 percent for those persons having at least an associate degree in engineering or science.

5. CP-189 requires experience records to:
 - a. be verified for initial and continuing experience.
 - b. only be addressed for initial certification.
 - c. only be needed if the individual holds a valid employer's Level III certificate.
 - d. be kept only for the initial certification period.
 - e. be updated monthly.
6. Per CP-189, the vision examinations shall consist of a
 - a. near distance acuity reading a Jaeger #2 chart at a distance of not less than 12 inches administered annually.
 - b. color vision examination which demonstrates the ability to differentiate among colors used in the method administered annually.
 - c. far distance vision test administered initially and at recertification.
 - d. a. and b. above.
 - e. none of the above.
7. Certification of Level III personnel per CP-189 requires the employer to administer the following examinations:
 - a. Basic, Method and Specific.
 - b. Vision, Basic, Method and Specific.
 - c. General, Specific and Practical.
 - d. Vision, Specific and Practical.
 - e. Specific and Practical
8. CP-189 requires the NDT Level III, responsible for administration and grading of NDT examinations, must be:
 - a. a holder of a currently valid ASNT Level III certificate in the method.
 - b. any Level III designated as the principal Level III.
 - c. any Level III appointed by the employer.
 - d. a Level III employed by the same company.
 - e. an outside agency Level III.
9. Level III certifications per CP-189 shall expire when:
 - a. employment with the employer is terminated.
 - b. the vision examination interval exceeds one year.
 - c. the ASNT Level III certificate has expired.
 - d. a. or b. above.
 - e. a. or c. above.

10. According to SNT-TC-1A, an NDT Level III individual should:
- be knowledgeable about all international as well as domestic certification programs.
 - be capable of developing, qualifying, and approving procedures.
 - also be certified to at least Level II in another method.
 - have at least 5 years of experience in the NDT method(s) in which certified.
 - both b. and d. above.
11. When certifying directly to UT Level II per CP-189, the minimum number of questions for the Specific exam would be:
- 20 questions, with a grade of at least 80%.
 - 20 questions, with a grade of at least 70%.
 - 40 questions, with a grade of at least 80%.
 - 40 questions, with a grade of at least 70%.
 - 30 questions.
12. Which action by an ASNT certified Level III would be considered as a violation of Level III Code of Ethics?
- Use of a brochure describing their ASNT certifications for the purposes of soliciting employment.
 - Reporting the felony conviction of another ASNT Level III while his/her Level III certification is valid.
 - Refusing to accept gratuities from the employer's client.
 - Expressing an opinion as an expert witness in a court trial concerning matters in which the Level III is certified.
 - Certifying NDT personnel without adequate knowledge of their capabilities.
13. Organized NDT training of personnel in accordance with CP-189 shall be presented by an individual qualified as:
- NDT Level III
 - NDT Level II
 - Metallurgist or welding engineer
 - NDT instructor
 - ASNT Level III.
14. The recommendations in SNT-TC-1A for leak testing:
- differ in minimum training hours for the four techniques.
 - do not consider different techniques.
 - are listed for Levels II and III only.
 - are significantly greater than for ultrasonic testing.
 - do not include pressure change testing.

15. The levels of qualification specified in CP-189 are:
- trainee, Level I, Level II and Level III.
 - NDT instructor, trainee, Level I, Level II, Level II Limited, and Level III.
 - Level I, Level II and Level III.
 - NDT instructor, Level I, Level II, and Level III
16. CP-189 requires that the employer records include:
- copies of current and previous employer examinations.
 - a monthly log of experience.
 - a copy of high school diploma or equivalent.
 - the NDT method, test technique, and level of certification.
 - all of the above.
17. Per CP-189, the employer's certification procedure shall be approved by the:
- employer's representative designated by the NDT Level III.
 - customer.
 - NDT Level III designated by the employer.
 - head of ASNT Technical Services.
 - employer's Quality Assurance Manager.
18. Per CP-189, for an individual to receive credit for NDT training that individual must:
- have attended the entire course.
 - have a passing composite score.
 - pass all quizzes and examinations.
 - pass a final examination.
19. Which of the following may conduct and grade all qualification examinations (general, specific and practical) for Level I and Level II personnel per CP-189?
- An NDT Level III in that method.
 - A selected representative of the NDT Level III.
 - An ASNT Level III certificate holder in that method.
 - An NDT Level II if designated as a qualified representative.
20. Which of the following would not be required by CP-189?
- Minimum required training hours for each NDT method/technique.
 - Periodic technical performance evaluation.
 - Annual vision exam for near distance acuity.
 - For reexamination after a failed exam, receive additional training or wait at least 30 days.

SECTION II: NDT METHODS

Radiographic Testing

21. The difference between the densities of two areas of a radiograph is called:
- a. radiographic contrast.
 - b. subject contrast.
 - c. film contrast.
 - d. definition.
 - e. opacity.
22. The penetrating ability of an x-ray beam is governed by:
- a. kilovoltage or wavelength.
 - b. time.
 - c. milliamperage.
 - d. source-to-film distance.
23. The fact that gases, when bombarded by radiation, ionize and become electrical conductors make them useful in:
- a. x-ray transformers.
 - b. x-ray tubes.
 - c. masks.
 - d. radiation detection equipment.
 - e. fluoroscopy.
24. An unshielded isotope source gives a dosage rate of 900 mR/hour at 10 feet. What would the unshielded dosage rate be at 30 feet? $I_1 / I_2 = D_2^2 / D_1^2$
- a. 300 mR/hr
 - b. 600 mR/hr
 - c. 100 mR/hr
 - d. 2700 mR/hr
25. Radiation intensity at 6 feet from an isotope is 40 R/hr. At what distance would the intensity be reduced to 10 R/hr?
- a. 10 feet
 - b. 12 feet
 - c. 20 feet
 - d. 144 feet
 - e. 3 feet

$$D_2 = \sqrt{\frac{I_1 \times D_1^2}{I_2}}$$

26. Radiation intensity at a certain point is 20 R/hr. How many HVL are required to reduce the intensity to 5 R/hr?

- a. 2 HVL
- b. 4 HVL
- c. 5 HVL
- d. 10 HVL
- e. 20 HVL

27. Which of the following technique variables is most commonly used to adjust subject contrast?

- a. Source-film distance
- b. Milliamperage
- c. Kilovoltage
- d. Focal spot size
- e. All of the above.

28. A penetrometer is used to measure the:

- a. size of discontinuities in a part.
- b. density of the film.
- c. amount of film contrast.
- d. quality of the radiographic technique.
- e. amount of radiation that penetrates the test object.

29. When viewing a radiograph, an image of the back of the cassette superimposed on the image of the specimen is noted. This is most likely due to:

- a. undercut.
- b. over-exposure.
- c. x-ray intensity too high.
- d. back-scattered radiation.
- e. under exposure

Magnetic Particle Testing

30. During a head shot, circular magnetization is useful in the detection of:

- a. circumferential cracks.
- b. longitudinal cracks.
- c. cracks in cylindrical parts at right angles to the long axis of the part.
- d. both a. and c. above.
- e. surface indications only

31. Using the rule of thumb formula for amperage calculation at 300-800 amps/inch of diameter, what amperage would be used to circularly magnetize a part that is 2.75" diameter and 13.5" long?
- 1575 amps
 - 2275 amps
 - 2750 amps
 - 3920 amps
32. Referring to question 31, what amperage would be required if the part is placed against the inside surface of a 12" diameter, 5 turn coil? $NI = 45,000 / (L/D) [+/- 10\%]$
- About 2100 amps
 - About 1553 amps
 - About 1830 amps
 - About 9183 amps
33. How is the inside diameter of a cylinder best magnetized?
- By a head shot.
 - By using prods at either end.
 - With a central conductor placed between contact heads.
 - With the cylinder placed crosswise in a solenoid.
34. What amperage would be used for MT inspecting a butt weld of 3/4-inch plates using contact prods and 7" prod spacing using the rule 90 to 100 amps/inch:
- 700-875 amps
 - 725-875 amps
 - 825 amps
 - 630-700 amps
 - none of the above
35. Which of the following is a major disadvantage of using the prod method?
- The magnetic field may be applied in the wrong direction.
 - The inspection surface may be arc burned.
 - Magnetic saturation may occur.
 - The operator's eyes may be burned by arc strikes.
36. The best type of magnetizing current for inspection of fatigue cracks is:
- direct current.
 - alternating current.
 - half-wave direct current.
 - full-wave rectified direct current.
 - any of the above if the amperage is correct.

37. Fluorescent magnetic particles are used in preference to visible magnetic particles:

- a. when parts are big and bulky.
- b. to take advantage of the fluorescent lights that are normally used in plants.
- c. if parts are for railroad applications.
- d. to increase the speed and reliability of detecting very small discontinuities.

38. Which of the following materials cannot be tested by magnetic particle techniques?

- a. High alloy steels
- b. Tool steels
- c. Copper alloys
- d. Ferritic stainless steels
- e. Both c. and d. above

Magnetic Flux Leakage

39. The current used for magnetization when using magnetic flux leakage inspection must be a:

- a. current that reverses direction, such as alternating current.
- b. steady non-fluctuating current.
- c. current that varies based on material thickness.
- d. current that fluctuates on and off at a consistent rate.

40. When inspecting wire rope, a magnetic flux loop is used to monitor:

- a. broken external wires.
- b. broken internal wires.
- c. metal loss from corrosion.
- d. all of the above.

41. In magnetic flux leakage inspection for discontinuities using an active field, the part being inspected should be magnetized:

- a. to the point of maximum permeability.
- b. well below saturation.
- c. to saturation or near saturation.
- d. to the point of maximum permeability.

42. Magnetism applied to a ferromagnetic part by a permanent magnet is called:

- a. induced magnetization.
- b. direct magnetization.
- c. direct current magnetization.
- d. permanent magnetization.

43. What particular type of discontinuity would not typically be indicated by magnetic flux leakage techniques?

- a. Fabrication scar.
- b. Surface contamination.
- c. Longitudinal seam.
- d. Pitting with cracking.

Ultrasonic Testing

44. Ultrasonic testing techniques are useful in testing laminate and sandwich construction test objects for:

- a. paint thickness.
- b. bond integrity.
- c. leakage.
- d. surface roughness.
- e. electrical conductivity.

45. An ultrasonic test using a straight-beam contact search unit is being conducted through the thickness of a flat part such as a plate. The test should detect:

- a. laminar type flaws with major dimension parallel to the plane of the rolled surface.
- b. transverse type flaws with major dimensions at right angles to the plane of the rolled surface.
- c. radial flaws with major dimensions along the length, but radially oriented to the rolled surface.
- d. none of the above.

46. Generally, the best ultrasonic testing technique for detecting discontinuities oriented along the fusion zone in a welded plate is a(n):

- a. angle-beam contact method employing surface waves.
- b. contact test using a straight longitudinal wave.
- c. immersion test using surface waves.
- d. resonance technique.
- e. angle-beam method using shear waves.

47. Ultrasonic inspection of castings is occasionally impractical because of:

- a. extremely small grain structure
- b. coarse grain structure
- c. uniform flow lines
- d. random orientation of discontinuities

48. The display on the majority of pulse-echo ultrasonic instruments consists of:

- a. automatic read-out equipment.
- b. an A-scan presentation.
- c. a B-scan presentation.
- d. a C-scan presentation.

49. Lack of parallelism between the entry surface and the back surface:

- a. may result in a screen pattern that does not contain back reflection indications.
- b. makes it difficult to locate discontinuities that lie parallel to the entry surface.
- c. usually indicates a porous condition existing in the metal.
- d. will decrease the penetrating power of the test.

50. The amount of energy reflected from a discontinuity is dependent on:

- a. the size of the discontinuity.
- b. the orientation of the discontinuity.
- c. the type of the discontinuity.
- d. all of the above.
- e. only a. and b. above.

51. The most commonly used method of producing shear waves in a test part when inspecting by the immersion method is by:

- a. transmitting longitudinal waves into a part perpendicular to its front surface.
- b. using two crystals vibrating at different frequencies.
- c. using a low frequency transducer.
- d. angulating the search tube to the proper angle with respect to the entry surface of the test part.

52. Ultrasonic velocities vary for different materials. These variances are primarily caused by differences in the material's:

- a. frequency and wavelength.
- b. thickness and travel time.
- c. density and elasticity.
- d. chemistry and permeability.

Penetrant Testing

53. Which of the following is not a characteristic that applies to penetrant testing?

- a. Method can accurately measure the depth of a crack or discontinuity.
- b. Method can be used for on-site testing of large parts.
- c. Method can be used to find shallow surface discontinuities.
- d. Method can be made more or less sensitive by using different penetrant materials.

54. The property of a dye used in penetrant materials to emit light of a range of wavelengths different from the wavelength of light that excites the emission is called:

- a. emissivity.
- b. irradiation.
- c. blacklight.
- d. spectrum blocking.
- e. fluorescence.

55. Penetrant testing is a nondestructive test which can be used for:

- a. locating and evaluating all types of discontinuities in a test specimen.
- b. locating and determining the length, width, and depth of discontinuities in a test specimen.
- c. determining the tensile strength of a test specimen.
- d. locating discontinuities open to the surface.
- e. all but c. above.

56. A red penetrant indication against a white background is most likely to be seen when:

- a. dry developers are used.
- b. visible dye penetrants are used.
- c. fluorescent post emulsified penetrants are used.
- d. wet developers are used.

57. A method used for comparing the overall performance of an in-use penetrant material is to:

- a. determine the viscosity of the penetrant.
- b. use a specially cracked aluminum panel.
- c. compare two sections of an aluminum panel with no discontinuities.
- d. determine the wettability of the penetrant.

58. When using a solvent removable penetrant, which of the following is normally considered the proper step(s) for removing the excess penetrant from the surface?

- a. Squirting solvent over the surface with no more than 10 psi pressure.
- b. Wiping with a solvent soaked cloth, then wiping with a dry cloth.
- c. Wiping with a solvent dampened cloth, then wiping with dry cloths.
- d. Wiping with a dry wipe, followed by wiping with a solvent dampened cloth, and then allowing the surface to dry.

59. In a penetrant test, scattered round indications on the part surface could be indicative of:

- a. fatigue cracks.
- b. porosity.
- c. weld laps.
- d. hot tears.
- e. grinding cracks.

60. Which of the following statements does not apply to developers used during penetrant testing?
- a. Developers are normally high fluorescent.
 - b. Some developers furnish a contrasting background during inspection.
 - c. Developers absorb or blot the penetrant that remains in discontinuities after the excess penetrant has been removed.
 - d. Both b. and c. above.

61. The function of emulsifier in the post-emulsified penetrant process is to:
- a. more rapidly drive the penetrant into deep, tight cracks.
 - b. react with the surface penetrant to make the penetrant water-washable.
 - c. add fluorescent dye or pigment to the penetrant.
 - d. provide a coating to which dry powder developer can adhere.
 - e. emulsify surface oils and greases to facilitate their removal.

Electromagnetic Testing

62. Eddy current testing relies on the principle of:
- a. magnetostriction.
 - b. electromagnetic induction.
 - c. piezoelectric energy conversion.
 - d. magnetomotive force.
 - e. direct current flow in a resistive circuit.
63. When the electrical current in an electromagnetic coil reverses direction, the:
- a. direction of the eddy currents in a part remains the same.
 - b. eddy currents in the test part will change phase by 45°.
 - c. direction of the eddy currents in the part also reverses.
 - d. eddy currents in the part remain the same.
64. In order to generate measurable eddy currents in a test specimen, the specimen must be:
- a. an electrical conductor.
 - b. an electrical insulator.
 - c. either a conductor or insulator.
 - d. a ferromagnetic material.
 - e. a non-magnetic material.

65. Eddy currents are circulating electrical currents induced in conductive materials by:

- a. continuous direct current
- b. alpha rays
- c. an alternating magnetic field
- d. a piezoelectric force
- e. any of the above

66. A term used to describe the effect observed due to change in the coupling between a test specimen and a flat probe coil when the distance of separation between them is varied is:

- a. fill factor.
- b. edge effect.
- c. end effect.
- d. lift-off.
- e. coupling loss.

67. In electromagnetic testing, IACS is a recognized abbreviation for:

- a. induced alternating current system.
- b. inductively activated comparison system.
- c. internal applied current system.
- d. international annealed copper standard.

68. The term "fill factor" applies to:

- a. a surface coil.
- b. a coaxial cable.
- c. an encircling coil.
- d. the ability to null an eddy current instrument.

69. The thickness of nonconductive coatings on a conductive base can be most simply measured by:

- a. observing the lift-off effect caused by the coating.
- b. testing both sides of the specimen.
- c. varying the test frequency over a given range during the test.
- d. using a specially shaped encircling coil.

70. It is often possible to sort various alloys of a nonmagnetic metal by means of an electromagnetic test when:

- a. there is a unique range of permeability values for each alloy.
- b. there is a unique range of conductivity values for each alloy.
- c. the direction of induced eddy currents varies for each alloy.
- d. the magnetic domains for each alloy are different.
- e. the alloy is slightly magnetic, but not if totally non-magnetic.

Neutron Radiography Testing

71. Which of the following neutron radiography converter foils cannot be used for transfer or indirect radiography?

- a. Dysprosium.
- b. Indium.
- c. Gadolinium.
- d. Gold

72. A thin sheet of gadolinium foil, in intimate contact with photographic film during neutron exposure, will increase film density because it:

- a. fluoresces and emits visible light, which helps expose the film.
- b. emits electrons that darken the film.
- c. absorbs the scattered radiation.
- d. prevents back scattered radiation from fogging the film.

73. Neutron radiography cannot be used for inspecting which of the following applications?

- a. Presence of explosive in a metal device.
- b. Presence of foreign materials such as oils.
- c. Welds in high-strength steels.
- d. Lubricants in metal systems.
- e. Nuclear fuel elements.

74. A radioactive isotope used for neutron radiography is:

- a. Cf-252
- b. Pu-239
- c. Ir-192
- d. Se -75

75. All of the following are neutron imaging detectors except:

- a. gadolinium convertor with x-ray film.
- b. conventional X-ray film.
- c. scintillator, light-emitting fluorescent screen.
- d. dysprosium or indium foil activation transferred to film.

Leak Testing

76. Of all the LT techniques, which of the following is an advantage of the bubble leak test?
- The results are quantitative in nature.
 - Limited skill and training is required for personnel.
 - Very high sensitivity is attainable.
 - It is not satisfactory for detecting gross leakage.
77. A helium mass spectrometer is a leak testing instrument that can be used with the:
- detector probe mode.
 - tracer probe mode.
 - bell jar mode.
 - all of the above.
78. The passage of gas by diffusion through a solid material where there is no physical leak in the sense of an opening or defect is called:
- absorption.
 - adsorption.
 - capillary action.
 - permeation.
79. What are three essential parameters involved in the dynamics of gases when leak testing?
- Volume, temperature and pressure.
 - Volume, temperature and density.
 - Density, atomic weight and pressure.
 - Pressure, specific gravity and atomic weight.
80. A virtual leak is:
- a leak that is small enough to be within the acceptance standard and therefore does not have to be repaired.
 - a leak from an isolated pocket within a vacuum system.
 - a leak that is so small as to be virtually impossible to detect.
 - outgassing vapors within a vacuum system.
81. Safe test practices dictate that all gages used for vacuum box bubble testing should be calibrated:
- at least every 2 weeks.
 - when taken from storage and prior to using.
 - once a month whether in use or in storage.
 - none of the above.

82. Gages used for vacuum box bubble leak testing should preferably have dials graduated over a range:
- more than 5 times the intended pressure.
 - of atmospheric pressure.
 - in degrees Celsius.
 - 3.6 times the maximum intended pressure.
83. Which one of the following elements is detectable with a halogen diode detector:
- helium.
 - argon.
 - carbon dioxide.
 - chlorine.
84. Most leaks detected during a detector-probe (sniffer) test could have been detected, and usually can be verified by:
- a bubble leak test
 - an ultrasonic examination
 - a visual examination
 - a pressure change measurement test

Acoustic Emission Testing

85. Attenuation of a wave is best defined by which of the following statements?
- Decrease in frequency with the distance traveled.
 - Decrease in amplitude with distance traveled.
 - Decrease in wave speed with distance traveled.
 - Change in direction with a function of time.
86. Which of the following factors will tend to produce low amplitude acoustic emission response during a tensile test?
- Low temperature.
 - High strain rate.
 - Plastic deformation.
 - Crack propagation.
87. Acoustic emission transducers are piezoelectric sensors which usually operate at frequencies from:
- 1 MHz - 5 MHz
 - 100 Hz - 2 kHz
 - 100 kHz - 300 kHz
 - 500 kHz - 0.5 MHz

88. The Felicity ratio is a quantitative measure used to evaluate:

- a. carbon steel reactors.
- b. fiberglass vessels.
- c. stainless steel piping.
- d. aluminum aircraft structures.

89. A typical source mechanism of acoustic emission is:

- a. crack growth.
- b. movement of dislocations.
- c. matrix cracking in fiber-reinforced plastics.
- d. all of the above.

90. The positioning of sensors for AE testing of metal pressure vessels is commonly based on:

- a. the measured attenuation in the structure.
- b. the need to detect structural flaws at critical locations.
- c. the velocity of sound in the structure.
- d. both a. and b. above.

91. Of the following components, which one is not considered to be part of a typical acoustic emission sensor:

- a. damping material.
- b. magneto coil.
- c. piezoelectric element.
- d. electrodes.
- e. cable connector.

92. When performing AE source location, which of the following most directly affects the accuracy of computed location:

- a. accuracy of sensor placement.
- b. physical size of a sensor.
- c. sensor frequency.
- d. sensor couplant.

93. When selecting the best sensor frequency for a particular acoustic emission test, it is important to consider all of the following except:

- a. attenuation characteristics of the material.
- b. frequency spectrum and level of background noise.
- c. cable length.
- d. sensor spacing.

Thermal / Infrared Testing

94. Which of the following conditions would be detected with thermography?
- a. Unbond conditions.
 - b. Presence of moisture in structures.
 - c. Heat loss from components.
 - d. Leaks in flat, composite roofs.
 - e. All of the above.

95. Thermography sensors are capable of detecting electromagnetic radiation wavelengths:
- a. shorter than the gamma ray spectrum.
 - b. in the visible light spectrum.
 - c. longer than the radio and television wave spectrum.
 - d. in the infrared spectrum.

96. Which of the following would be considered to be a contact type sensor?

- a. Spot radiometer.
- b. Line scanner.
- c. Thermally quenched phosphor.
- d. Pyrometer.

97. Conduction, convection, and radiation all transfer heat energy. What drives the energy in all cases?

- a. Pressure difference.
- b. Emission force.
- c. Energy wavelength delta.
- d. Temperature difference.

98. With respect to steady state heat transfer by thermal conduction, which statement below is true?

$$Q = (k/l) \times A \times \Delta T$$

- a. Heat flux is directly proportional to the wall thickness and inversely proportional to the temperature difference.
- b. Heat flux is inversely proportional to the wall thickness and directly proportional to the temperature difference.
- c. Heat flux is directly proportional to the wall thickness and temperature difference.
- d. Heat flux is not related to the wall thickness or temperature.

99. Infrared cameras can record short wave spectral responses in the infrared region between ____.

- a. 0.5 to 0.7 micrometers
- b. 0.7 to 0.9 micrometers
- c. 3.0 to 5.0 micrometers
- d. 8.0 to 12.0 micrometers

Visual Testing

100. What is the name of the condition of eye adaptation to a dark environment?
- Photopic vision
 - Scotopic vision
 - Stereoscopic vision
 - Heliotopic vision
101. Luminous intensity:
- varies with the distance from the source.
 - varies with the distance from the square of the source.
 - is constant with the distance from the source.
 - is inversely proportional to the square of the distance from the source.
102. Visual acuity is the ability of the eye to:
- resolve fine details.
 - perceive light intensity.
 - distinguish between different colors.
 - retain images after the light source is removed.
 - identify specific wavelengths.
103. An optical device consisting of a tube with an eyepiece on one end, a light source, and an objective lens to illuminate and magnify a remote object is a:
- microscope
 - borescope
 - fiber-optic scope
 - telescope
104. The root of the weld refers to the:
- deepest point of useful weld penetration into a joint.
 - portion of the joint where the members approach closest to each other.
 - separation between members to be joined.
 - second welding pass made.

105. Which of the following weld symbols represents a single-bevel weld on the arrow side?

- a. 
- b. 
- c. 
- d. 

SECTION III: MATERIALS & PROCESSES

106. Which of the following statements is correct?

- a. Discontinuities can propagate and become defects.
- b. All discontinuities are defects.
- c. All defects will lead to failure if undetected; discontinuities can be harmless.
- d. All defects are internal flaws originating from errors in processing.

107. Most metals and plastics that are solids and have reasonable strength at room temperature are called:

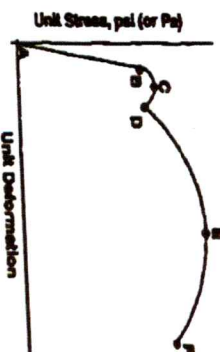
- a. raw materials.
- b. manufacturing materials.
- c. allotropic materials.
- d. engineering materials.

108. Tensile tests are conducted on specimens from a newly developed alloy in order to determine the ultimate tensile strength of the material. Such tests are referred to as:

- a. indirect tests.
- b. physical properties tests.
- c. destructive tests.
- d. proof tests.

109. On the stress-strain diagram below, point E is called the:

- a. elastic limit.
- b. ultimate strength.
- c. yield point.
- d. yield strength.
- e. modulus of elasticity.



110. Which of the following tests utilizes a pendulum to break a specimen that is notched and supported on both ends, with the result of measuring energy absorption?
- Creep test
 - Charpy test
 - Fatigue test
 - Transverse rupture test
111. The terms "body-centered cubic", "face-centered cubic", and "hexagonal closed-packed" all refer to the:
- different size grains that can exist at the same time in a metallic structure.
 - sequence of crystalline growth in a typical mild steel.
 - lattice structures that make up unit cells in a solid metallic structure.
 - change in a metallic structure as it undergoes plastic deformation.
112. In the iron and steel making process, the term "pig iron" refers to:
- the waste material that contains high concentrations of impurities and slag and is either discarded or used as a by-product.
 - a high carbon, low ductility metal that is produced in the blast furnace which can be used to make subsequent types of iron and steel.
 - the molten metal from the blast furnace that is not usable and is poured off into a series of crude castings called "pigs".
 - low cost metal used in large production factories.
113. The process of returning ductility to a cold-worked low carbon steel is called:
- precipitation.
 - recrystallization.
 - allotropic change.
 - austenization.
114. Austenitic stainless steels are paramagnetic. This means that:
- alternating current must be used when using the magnetic particle testing method.
 - the steel is very dense and, relative to other steels, difficult to penetrate with x-rays.
 - ultrasonics is the logical NDT method to choose because of the coarse-grained nature of a paramagnetic material.
 - the material has a very low permeability.

115. The heat treatment of aluminum for the purpose of hardening and strengthening:
- is not possible with aluminum alloys as they contain no carbon and cannot undergo allotropic changes.
 - can produce tensile strengths equivalent to some carbon steels.
 - requires the use of special furnaces and is rarely done as a practical application.
 - requires that iron and carbon be alloyed for the best results.
116. Manufacturing is a term that refers to processing that starts with raw material in a bulk form and is concerned mainly with processing the raw material in a manner that changes its:
- shape.
 - chemical form.
 - mechanical properties.
 - physical properties.
117. Which of the following is true regarding solidification of molten metal in a casting mold?
- The metal cools at a constant rate, thus providing fine equiaxed grains throughout.
 - Cooling takes place in phases having different rates that produce different types of grain structure in different sections of the casting.
 - Solidification occurs at a constant rate, beginning at the interior of the casting and progressing outward.
 - Thin sections tend to cool more rapidly than thin sections because thin sections consist mostly of fine equiaxed grains.
118. Another term for precision casting and the lost-wax process is:
- investment casting.
 - die casting.
 - metal mold casting.
 - shell mold casting.
119. During the solidification of a casting, the shrinkage that occurs:
- may cause porosity and shrinkage cavities primarily in the outer surfaces where the metal cools first.
 - requires that the pattern used be slightly smaller than the desired dimension of the finished casting.
 - may cause cavities that are enlarged by the evolution of gases.
 - may be eliminated by investment casting.

120. Small metal supports used to support and position cores become part of a casting by fusing with the molten metal. Such devices are called:

- a. core hangers.
- b. chills.
- c. chaplets.
- d. risers.
- e. patterns.

121. The part of the casting where the gate or riser was attached:

- a. should be the area used to establish reference standards for future NDT examinations because this area will contain the best quality cast material.
- b. provides the best quality material because of rapid cooling in this area.
- c. should be in the area that "freezes" first, thus preventing excessive shrinkage cavities.
- d. may provide a concentration point for discontinuities.

122. The uneven shrinkage and brittle structures that occur due to the rapid cooling of a weld can often be reduced by:

- a. preheating prior to welding.
- b. using a filler metal with a higher carbon content than the base metal.
- c. clamping the weldment in a rigid fixture.
- d. overdesigning the size of the weldment to prevent shrinkage.

123. Of the following, which is not a requirement for fusion bonding?

- a. Melting
- b. Atomic closeness
- c. Atomic cleanliness
- d. Pressure
- e. Both b. and c. above.

124. Which of the following welding processes uses a non-consumable electrode with the arc maintained in an atmosphere of inert gas?

- a. Gas tungsten arc welding.
- b. Submerged arc welding.
- c. Gas metal arc welding.
- d. Electroslag welding.

125. In arc welding, the electric arc is usually sustained between an electrode and the:

- a. electrode holder.
- b. coating on the electrode.
- c. workpiece.
- d. shielding gas.

126. Undercut on a weld pass is usually caused by:

- a. poor operator technique.
- b. too slow a rate of travel, causing the base metal to become too hot.
- c. using an electrode that is too large for the current capacity of the welding machine.
- d. welding in the vertical position.

127. Among other things, the advantageous effects of recrystallization depend upon the:

- a. rate of heating.
- b. temperature at which deformation takes place.
- c. presence of carbon in excess of 2.5% for steels.
- d. presence of silicon in excess of 0.1% for steels.

128. Which of the following would have the least ductility?

- a. Cold-rolled steel plate
- b. Hot-rolled steel plate
- c. Gray cast iron
- d. Hot-rolled aluminum plate

129. Slabs, blooms, and billets are:

- a. the three consecutive stages that the metal goes through during the production of products such as angle iron and channel iron.
- b. the shapes that the ingot is rolled into prior to a variety of secondary operations.
- c. types of defects that occur during the hot rolling of steel.
- d. the three different shapes produced during typical cold rolling operations.

130. Which of the following forms is generally selected for high strength and controlled property directionality?

- a. Castings
- b. Extrusions
- c. Hot-rolled bar stock
- d. Forgings

131. Powder metallurgy provides two unique advantages in metals processing. One is the capability to produce shapes and objects of refractory metals that are extremely difficult or impractical to melt. The other is to:

- a. economically produce metals with extremely low melting temperatures.
- b. produce metal shapes with controlled porosity.
- c. produce metals that can be easily machined by electrochemical processes.
- d. produce metals that are corrosion resistant.

132. Which of the following terms does not describe operations that bend, stretch and shape sheet metals into three-dimensional shapes with significant plastic flow and deformation?

- a. Bending
- b. Forming
- c. Drawing
- d. Shearing

133. The major difference between materials classified as composites and those classified as mixtures is that:

- a. mixtures are a type of composite with random orientation and shape of the constituents.
- b. composites contain metallic constituents and mixtures are nonmetallic.
- c. mixtures start as liquids blended together and composites start as solids.
- d. mixtures are elastomeric, while composites are characterized as having at least one plastic component.

134. The process used for shaping metals by chemical dissolution only, with selective removal accomplished by masking areas where metal is not to be removed, is called:

- a. electrical discharge machining.
- b. chemical milling.
- c. electrochemical machining.
- d. electroforming.
- e. none of the above.

135. Which of the following would make use of interferometry?

- a. Optical flats.
- b. Optical comparator.
- c. Microscope.
- d. X-ray fluorescence.
- e. Alternating current field measurement.

